

CSE 318: Artificial Intelligence Sessional

Online 01

C1

Non Uniform Cost

Problem Statement

Modify your existing N-Puzzle solver so that each move has a non-uniform and path-dependent cost.

In the original puzzle, every move had cost 1. In this task, moving a tile has a cost based on the face value of the tile and the number of times that tile has already been moved.

The blank tile 0 does not have any movement cost. The cost of a move is determined by the tile that slides into the blank position.

Tile Fatigue Rule

The first time a tile is moved, its cost is equal to its face value.

The second time the same tile is moved, its cost becomes twice its face value.

The third time the same tile is moved, its cost becomes three times its face value, and so on.

For example, for tile 7:

- First movement of tile 7: cost 7
- Second movement of tile 7: cost 14
- Third movement of tile 7: cost 21

Therefore, the cost of moving a tile depends on how many times that tile has already been moved in the current solution path.

Required Modification

Update your A* search so that $g(n)$ represents the **total path cost** from the initial board to the current board.

Heuristic Function

Use the weighted Manhattan distance as the heuristic:

$$h(n) = \sum_{\text{tiles } t} t \times (|r_t - r_t^*| + |c_t - c_t^*|)$$

Here, (r_t, c_t) is the current position of tile t , and (r_t^*, c_t^*) is the goal position of tile t .

The blank tile 0 should not be included in the heuristic calculation.

For the heuristic, you may ignore the tile fatigue cost. The fatigue rule only needs to be applied when updating $g(n)$.

Output

Your program should print the minimum total cost.

Then print the solution path from the initial board to the goal board, along with the total cost for each step.

If the puzzle is unsolvable, print:

`Unsolvable puzzle`